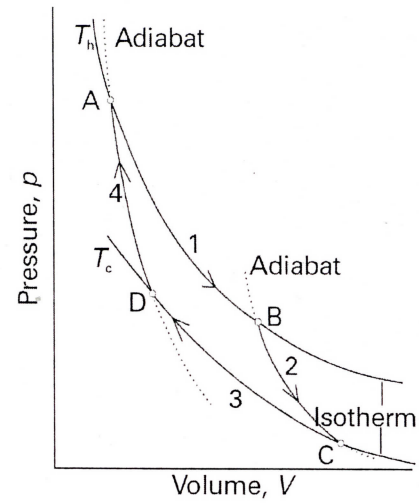
Physical Chemistry CUME, November 16, 2018

Thermodynamics

1. [40 points] A monoatomic ideal gas expands adiabatically and reversibly. Starting from the first law of thermodynamics, find an expression for the pressure (p) as a function of the volume (V), the initial pressure (p_0), and the initial volume (V_0). Hint: Although the volume is not constant in this process, you may use $dU = C_V\, dT$ since internal energy is a state function.

2. [40 points] For a mole of monoatomic gas molecules involved in the following processes of a thermodynamic cycle, find the entropy change associated with each step of the process assuming that $V_B = 2V_A$. Express your answers numerically in SI units.

   Step 1: A reversible, isothermal process from (p_A, V_A, T_h) to (p_B, V_B, T_h).
   Step 2: A reversible, adiabatic process from (p_B, V_B, T_h) to (p_C, V_C, T_c).
   Step 3: A reversible, isothermal process from (p_C, V_C, T_c) to (p_D, V_D, T_c).
   Step 4: A reversible, adiabatic process from (p_D, V_D, T_c) to (p_A, V_A, T_h).



3. [20 points] The standard molar entropy of argon is given as a function of temperature (expressed in K) by:

$$(36.4 + 20.8 \ln T)\, JK^{-1}mol^{-1}$$

Calculate the change in Gibbs energy when 1.0 mole of argon is heated at constant pressure from $20°C$ to $60°C$.

Useful Information

Gas constant R = 8.314 J mol^{-1} K^{-1}

$$\int \frac{dx}{x} = \ln x \qquad \int \ln x\; dx = x \ln x - x$$